Задача 1: $X \in \mathbb{R}^{n\times n}$, $f(X) = \ln\det X$; $\nabla_X f = ?$

$g(X) := \det X$, $z(g) := \ln g$; $\nabla_X f = \nabla_X g \cdot z'$

$z' = \frac{1}{g}$; $\nabla_X g = \left(\frac{\partial g}{\partial X_{ij}}\right)_{i,j=1}^{n,n}$

Т. Лапласа о разложении определителя по строке i: $\frac{\partial g}{\partial X_{ij}} = \frac{\partial}{\partial X_{ij}}\left(\sum_{k=1}^{n} X_{ik} A_{ik}\right) = A_{ij}$

т.к. A_{ik} не содержит элементов из i-ой строки

$X^{-1} = \frac{1}{\det X} A^T$; $\nabla_X g = A = \det X \cdot (X^{-1})^T \Rightarrow$

$\nabla_X f = (X^{-1})^T$

Задача 2: $f(x) = x^T \exp(xx^T)x$, $x \in \mathbb{R}^n$, $\nabla_x f(x) = ?$

$\exp(xx^T) = \sum_{k=0}^{\infty} \frac{(xx^T)^k}{k!}$; $\prod_{i=1}^{k} xx^T = (x^T x)^{k-1} xx^T \Rightarrow$

$\exp(xx^T) = \sum_{k=0}^{\infty} \frac{(x^T x)^{k-1}}{k!} \cdot xx^T$

$\exp(xx^T)x = \sum_{k=0}^{\infty} \frac{(x^T x)^k}{k!} x = e^{x^T x} x$;

$\lrcorner\ s = x^T x \Rightarrow f = s e^s$ $\frac{df}{ds} = e^s(1+s)$; $ds = (x^T dx + dx^T x)$

$; ds = 2x^T dx \Rightarrow df = \frac{df}{ds} ds = 2e^s(1+s)x^T dx =$

$= tr(2(1+x^T x) e^{x^T x} x^T dx) \Rightarrow \nabla_x f = 2(1+x^T x)e^{x^T x} x$

Задача 3: $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m; x \in \mathbb{R}^n; \nabla_x f(x) = ?$

$f(x) = \sin\|Ax+b\|_2$; $g(x) := \|Ax+b\|_2 = (Ax+b)^T(Ax+b)$;

$z(g) = \sin g$; $\nabla_x f(x) = \nabla_x g(x) \cdot z'(g)$; $z'(g) = \cos(g)$

$dg(x) = (A\,dx)^T(Ax+b) + (Ax+b)A\,dx = 2(Ax+b)^T A\,dx$

$g(x)$ - скаляр $\Rightarrow dg(x) = tr(2(Ax+b)^T A\,dx) \Rightarrow$

$\nabla_x g(x) = 2((Ax+b)^T A)^T = 2A^T(Ax+b) \Rightarrow$

$\nabla_x f(x) = 2\cos\|Ax+b\|_2 A^T(Ax+b)$

Задача 4: $A \in \mathbb{R}^{n \times n}, A = Q\Lambda Q^T, \Lambda = I_n\lambda$;

$\nabla_\lambda tr(A) = ?$; $d\,tr(A) = d_\lambda tr(Q I_n \lambda Q^T) = tr(Q^T Q I_n d\lambda) \Rightarrow$

$\nabla_\lambda tr(A) = I_n Q^T Q$

$\nabla_Q tr(A) = ?$ $d_Q tr(A) = d_Q tr(Q^T Q I_n \lambda) =$

$\cancel{= tr(dQ^T Q I_n \lambda + Q^T d_Q I_n \lambda)}$

$\cancel{dQ^T}$ Q - ортогональная матрица $\Rightarrow Q^T Q = I_n \Rightarrow$

$d_Q tr(A) = d_Q tr(I_n \lambda) \Rightarrow \nabla_Q tr(A) = (0)_{i,j=1}^{n,n}$

Задача 5: $Q(w) = \frac{1}{\ell}\sum_{i=1}^{\ell} \ln(\cosh(w^T x_i - y_i))$

$\nabla_w Q(w) = ?$; $q_i(w) := \ln(\cosh(w^T x_i - y_i))$

$f(w) := w^T x_i - y_i$, $g(f) := \cosh f$; $z(g) = \ln g$

$\nabla_w q_i(w) = \nabla_w f(w) \cdot g'(f) \cdot z'(g)$;

$g'(f) = \frac{1}{2}(e^{f} + e^{-f})' = \frac{1}{2}(e^{f} - e^{-f}) = \sinh f$

$z'(g) = \frac{1}{g}$

$\nabla_w f(w) = \left(\left(\frac{\partial f}{\partial w_j}\right)_{j=1}^{d}\right)^T$; $\frac{\partial f}{\partial w_j} = x_j \Rightarrow \nabla_w f(w) = x_i$

$\nabla_w q_i(w) = tg(w^T x_i - y_i) x_i \Rightarrow$

$\nabla_w Q(w) = \frac{1}{\ell}\sum_{i=1}^{\ell} tg(w^T x_i - y_i) x_i = \frac{1}{\ell} X^T tg(wX - y)$

Задача 6: $(y - Xw)^T (y - Xw) + \lambda w^T w = Q(w)$

$Q(w)$ - скаляр $\Rightarrow dQ = tr((-X dw)^T (y - Xw) -$

$- (y - Xw)^T X dw + \lambda dw^T w + \lambda w^T dw =$

$= tr(2(\lambda w^T - (y - Xw)^T X) dw) \Rightarrow$

$\nabla_w Q = 2(\lambda w - X^T(y - Xw) = 0 \Rightarrow$

$(\lambda + X^T X) w = X^T y \Rightarrow w = \frac{X^T y}{\lambda + X^T X}$

$d \nabla_w Q = 2(\lambda + X^T X) dw \Rightarrow \nabla_w^2 Q = \lambda + X^T X$

$w^{(k)} = w^{(k-1)} - \eta \nabla_w Q = w^{(k-1)} - 2\eta(\lambda w^{(k-1)} - X^T(y - Xw^{(k-1)})$

Задание 7: $f(X) = tr((X-A)^T(X-A))$, $f(X) \to \min\limits_X$

$g(X) := X^T - X = 0$; $\Lambda \in \mathbb{R}^{n\times n}$

$\mathcal{L}(X,\Lambda) = f(X) - tr(\Lambda^T g(X))$ т.к. $tr(\Lambda^T g) = \sum_{i,j=1}^{n,n} \Lambda_{ij} g_{ij} \in \mathbb{R}$

$d_X\mathcal{L} = tr(dX^T(X-A) + (X-A)^T dX - \Lambda^T dX^T + \Lambda^T dX) =$

$= tr((2(X-A)^T - \Lambda + \Lambda^T)dX) \Rightarrow$

$\nabla_X \mathcal{L} = 2(X-A) - \Lambda^T + \Lambda = 0$

$\forall i,j:\ 2(X_{ij}-A_{ij}) - \Lambda_{ji} + \Lambda_{ij} = 0 \ (1)$ $\exists!\ 2(X_{ji}-A_{ji}) - \Lambda_{ij} + \Lambda_{ji} = 0 \ (2)$

$(1)+(2):\ X_{ij} + X_{ji} = A_{ij} + A_{ji}$; $X^T = X \Rightarrow$

$X = \dfrac{A + A^T}{2}$